

MOTOR AND GENERATOR MATH

$$V = E_g - I_A R_A$$

$$V = E_b + I_A R_A$$

Example 26.3. A shunt generator delivers 450 A at 230 V and the resistance of the shunt field and armature are $50\ \Omega$ and $0.03\ \Omega$ respectively. Calculate the generated e.m.f.

Solution. Generator circuit is shown in Fig. 26.46.

Current through shunt field winding is

$$I_{sh} = 230/50 = 4.6\text{ A}$$

Load current $I = 450\text{ A}$

$$\therefore \text{Armature current } I_a = I + I_{sh} \\ = 450 + 4.6 = 454.6\text{ A}$$

Armature voltage drop

$$I_a R_a = 454.6 \times 0.03 = 13.6\text{ V}$$

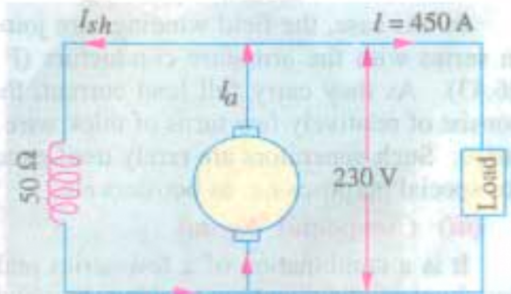


Fig. 26.46

Now $E_g = \text{terminal voltage} + \text{armature drop}$

$$= V + I_a R_a$$

$\therefore$ e.m.f. generated in the armature

$$E_g = 230 + 13.6 = 243.6\text{ V}$$

Example 26.4. A long-shunt compound generator delivers a load current of 50 A at 500 V and has armature, series field and shunt field resistances of $0.05\ \Omega$, $0.03\ \Omega$ and $250\ \Omega$ respectively. Calculate the generated voltage and the armature current. Allow 1 V per brush for contact drop.

(Elect. Science 1, Allahabad Univ. 1992)

Solution. Generator circuit is shown in Fig. 26.47.

$$I_{sh} = 500/250 = 2\text{ A}$$

Current through armature and series winding is

$$= 50 + 2 = 52\text{ A}$$

Voltage drop on series field winding

$$= 52 \times 0.03 = 1.56\text{ V}$$

Armature voltage drop

$$I_a R_a = 52 \times 0.05 = 2.6\text{ V}$$

Drop at brushes $= 2 \times 1 = 2\text{ V}$

Now, $E_g = V + I_a R_a + \text{series drop} + \text{brush drop}$

$$= 500 + 2.6 + 1.56 + 2 = 506.16\text{ V}$$

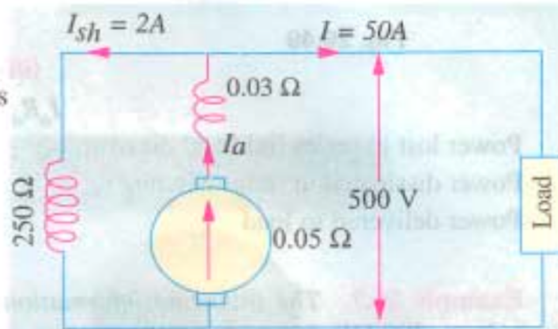


Fig. 26.47

Example 29.1. A 220-V d.c. machine has an armature resistance of 0.5Ω . If the full-load armature current is 20 A, find the induced e.m.f. when the machine acts as (i) generator (ii) motor.

(Electrical Technology-I, Bombay Univ. 1987)

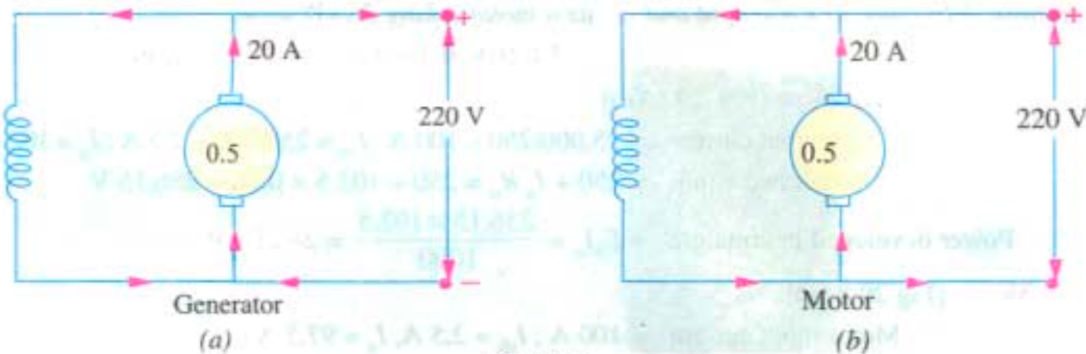


Fig. 29.7

Solution. As shown in Fig. 29.7, the d.c. machine is assumed to be shunt-connected. In each case, shunt current is considered negligible because its value is not given.

(a) As Generator [Fig. 29.7(a)] $E_g = V + I_a R_a = 220 + 0.5 \times 20 = 230 \text{ V}$

(b) As Motor [Fig 29.7 (b)] $E_b = V - I_a R_a = 220 - 0.5 \times 20 = 210 \text{ V}$

Example 29.3. A 440-V, shunt motor has armature resistance of 0.8Ω and field resistance of 200Ω . Determine the back e.m.f. when giving an output of 7.46 kW at 85 percent efficiency.

Solution. Motor input power = $7.46 \times 10^3 / 0.85 \text{ W}$

Motor input current = $7460 / 0.85 \times 440 = 19.95 \text{ A}$; $I_{sh} = 440 / 200 = 2.2 \text{ A}$

$I_a = 19.95 - 2.2 = 17.75 \text{ A}$; Now, $E_b = V - I_a R_a$

$E_b = 440 - (17.75 \times 0.8) = 425.8 \text{ V}$

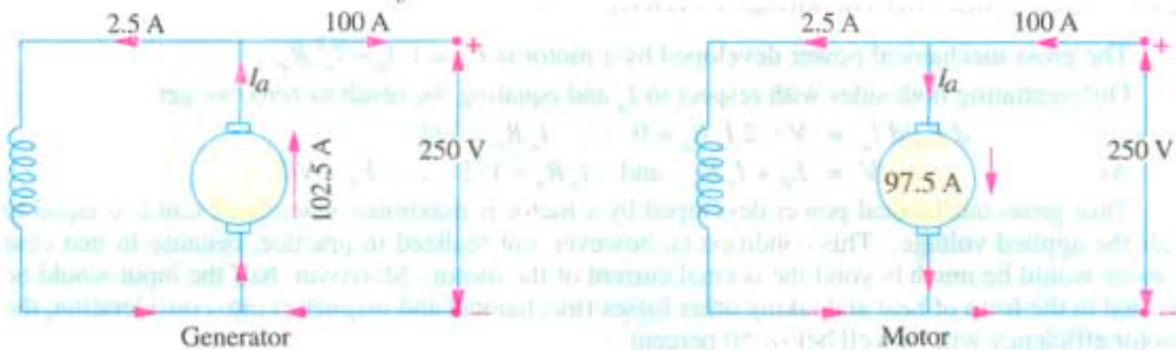


Fig. 29.8 (a)

Fig. 29.8 (b)

Transformer

Type: 1

① $\frac{E_2}{E_1} = \frac{N_2}{N_1} = \frac{I_1}{I_2} = K$, K is the transformation Ratio.

Mat R → Theraja →
 Ch → 32
 32.1 - 32.4

$N_2 > N_1 \rightarrow$ Step UP x-former
 $N_1 > N_2 \rightarrow$ Step down "

② $E_1 = 4.44 f N_1 \Phi_m$ $E_2 = 4.44 f N_2 B_m A$
 $= 4.44 f N_1 B_m A$

Primary

$$E_P = 4.44 f N_P B_m A$$

Secondary

$$E_S = 4.44 f N_S B_m A$$

Example 32.1. The maximum flux density in the core of a 250/3000-volts, 50-Hz single-phase transformer is 1.2 Wb/m^2 . If the e.m.f. per turn is 8 volt, determine

(i) primary and secondary turns (ii) area of the core.

(Electrical Engg.-I, Nagpur Univ. 1991)

Solution. (i)

$$E_1 = N_1 \times \text{e.m.f. induced/turn}$$

$$N_1 = 250/8 = 32; N_2 = 3000/8 = 375$$

(ii) We may use

$$E_2 = -4.44 f N_2 B_m A$$

$\therefore$

$$3000 = 4.44 \times 50 \times 375 \times 1.2 \times A; A = 0.03 \text{m}^2.$$

Example 32.3. A single-phase transformer has 400 primary and 1000 secondary turns. The net cross-sectional area of the core is 60 cm^2 . If the primary winding be connected to a 50-Hz supply at 520 V, calculate (i) the peak value of flux density in the core (ii) the voltage induced in the secondary winding. **(Elect. Engg-I, Pune Univ. 1989)**

Solution.

$$K = N_2/N_1 = 1000/400 = 2.5$$

(i)

$$E_2/E_1 = K \therefore E_2 = KE_1 = 2.5 \times 520 = \mathbf{1300 \text{ V}}$$

(ii)

$$E_1 = 4.44 f N_1 B_m A$$

or

$$520 = 4.44 \times 50 \times 400 \times B_m \times (60 \times 10^{-4}) \therefore B_m = \mathbf{0.976 \text{ Wb/m}^2}$$

Example 32.4. A 25-kVA transformer has 500 turns on the primary and 50 turns on the secondary winding. The primary is connected to 3000-V, 50-Hz supply. Find the full-load primary and secondary currents, the secondary e.m.f. and the maximum flux in the core. Neglect leakage drops and no-load primary current. **(Elect. & Electronic Engg., Madras Univ. 1985)**

Solution.

$$K = N_2/N_1 = 50/500 = 1/10$$

Now, full-load

$$I_1 = 25,000/3000 = \mathbf{8.33 \text{ A}}. \text{ F.L. } I_2 = I_1/K = 10 \times 8.33 = \mathbf{83.3 \text{ A}}$$

$$\text{e.m.f. per turn on primary side} = 3000/500 = 6 \text{ V}$$

$$\therefore \text{secondary e.m.f.} = 6 \times 50 = \mathbf{300 \text{ V}} \text{ (or } E_2 = KE_1 = 3000 \times 1/10 = 300 \text{ V)}$$

$$\text{Also, } E_1 = 4.44 f N_1 \Phi_m; 3000 = 4.44 \times 50 \times 500 \times \Phi_m \therefore \Phi_m = \mathbf{27 \text{ mWb}}$$

Type 2

Referred to primary $\rightarrow$

$$R_{01} = R_1 + R_2' \\ = R_1 + R_2/k^2$$

$$X_{01} = X_1 + X_2' \\ = X_1 + X_2/k^2$$

Impedance (total Resistance)

$$\Rightarrow Z_{01} = \sqrt{R_{01}^2 + X_{01}^2}$$

Referred to secondary $\rightarrow$

$$R_{02} = R_2 + R_1' \\ = R_2 + k^2 R_1$$

$$X_{02} = X_2 + X_1' \\ = X_2 + k^2 X_1$$

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2}$$

Example 32.15. A 30 kVA, 2400/120-V, 50-Hz transformer has a high voltage winding resistance of 0.1Ω and a leakage reactance of 0.22Ω . The low voltage winding resistance is 0.035Ω and the leakage reactance is 0.012Ω . Find the equivalent winding resistance, reactance and impedance referred to the (i) high voltage side and (ii) the low-voltage side.

(Electrical Machines-I, Bangalore Univ. 1987)

Solution.

$$K = 120/2400 = 1/20; R_1 = 0.1 \Omega, X_1 = 0.22 \Omega$$

$$R_2 = 0.035 \Omega \quad \text{and} \quad X_2 = 0.012 \Omega$$

(i) Here, high-voltage side is, obviously, the primary side. Hence, values as referred to primary side are

$$R_{01} = R_1 + R_2' = R_1 + R_2/K^2 = 0.1 + 0.035/(1/20)^2 = \mathbf{14.1 \Omega}$$

$$X_{01} = X_1 + X_2' = X_1 + X_2/K^2 = 0.22 + 0.012/(1/20)^2 = \mathbf{5.02 \Omega}$$

$$Z_{01} = \sqrt{R_{01}^2 + X_{01}^2} = \sqrt{14.1^2 + 5.02^2} = \mathbf{15 \Omega}$$

(ii)

$$R_{02} = R_2 + R_1' = R_2 + K^2 R_1 = 0.035 + (1/20)^2 \times 0.1 = \mathbf{0.03525 \Omega}$$

$$X_{02} = X_2 + X_1' = X_2 + K^2 X_1 = 0.012 + (1/20)^2 \times 0.22 = \mathbf{0.01255 \Omega}$$

$$Z_{02} = \sqrt{R_{02}^2 + X_{02}^2} = \sqrt{0.03525^2 + 0.01255^2} = \mathbf{0.0374 \Omega}$$

$$(\text{or } Z_{02} = K^2 Z_{01} = (1/20)^2 \times 15 = 0.0375 \Omega)$$

Example 32.16. A 50-kVA, 4,400/220-V transformer has $R_1 = 3.45 \Omega$, $R_2 = 0.009 \Omega$. The values of reactances are $X_1 = 5.2 \Omega$ and $X_2 = 0.015 \Omega$. Calculate for the transformer (i) equivalent resistance as referred to primary (ii) equivalent resistance as referred to secondary (iii) equivalent reactance as referred to both primary and secondary (iv) equivalent impedance as referred to both primary and secondary (v) total Cu loss, first using individual resistances of the two windings and secondly, using equivalent resistances as referred to each side.

(Elect. Engg.-I, Nagpur Univ. 1993)

Solution. Full-load $I_1 = 50,000/4,400 = 11.36 \text{ A}$ (assuming 100% efficiency)
 Full-load $I_2 = 50,000/220 = 227 \text{ A}; K = 220/4,400 = 1/20$

(i) $R_{01} = R_1 + \frac{R_2}{K^2} = 3.45 + \frac{0.009}{(1/20)^2} = 3.45 + 3.6 = 7.05 \Omega$

(ii) $R_{02} = R_2 + K^2 R_1 = 0.009 + (1/20)^2 \times 3.45 = 0.009 + 0.0086 = 0.0176 \Omega$
 Also, $R_{02} = K^2 R_{01} = (1/20)^2 \times 7.05 = 0.0176 \Omega$ (check)

(iii) $X_{01} = X_1 + X_2' = X_1 + X_2/K^2 = 5.2 + 0.015/(1/20)^2 = 11.2 \Omega$
 $X_{02} = X_2 + X_1' = X_2 + K^2 X_1 = 0.015 + 5.2/20^2 = 0.028 \Omega$
 Also $X_{02} = K^2 X_{01} = 11.2/400 = 0.028 \Omega$ (check)

(iv) $Z_{01} = \sqrt{(R_{01}^2 + X_{01}^2)} = \sqrt{(7.05^2 + 11.2)^2} = 13.23 \Omega$
 $Z_{02} = \sqrt{(R_{02}^2 + X_{02}^2)} = \sqrt{(0.0176^2 + 0.028)^2} = 0.03311 \Omega$
 Also $Z_{02} = K^2 Z_{01} = 13.23/400 = 0.0331 \Omega$ (check)

(v) Cu loss $= I_1^2 R_1 + I_2^2 R_2 = 11.36^2 \times 3.45 + 227^2 \times 0.009 = 910 \text{ W}$
 Also Cu loss $= I_1^2 R_{01} = 11.36^2 \times 7.05 = 910 \text{ W}$
 $= I_2^2 R_{02} = 227^2 \times 0.0176 = 910 \text{ W}$

Type: 3

Exact voltage drop $\rightarrow$

lagging power factor, $\rightarrow (I_2 R_{02} \cos \phi + I_2 X_{02} \sin \phi)$

leading " " $\rightarrow (I_2 R_{02} \cos \phi - I_2 X_{02} \sin \phi)$

Percentage drop $\rightarrow \left(V_R \cos \phi + V_X \sin \phi \right)$
 $\quad \quad \quad \downarrow \quad \quad \quad \downarrow$
 $\quad \quad \quad I_2 R_{02} \quad \quad I_2 X_{02}$

lagging $\rightarrow (V_R \cos \phi + V_X \sin \phi)$

leading $\rightarrow (v_r \cos \phi - v_x \sin \phi)$

Example 32.21. A 230/460-V transformer has a primary resistance of $0.2\ \Omega$ and reactance of $0.5\ \Omega$ and the corresponding values for the secondary are $0.75\ \Omega$ and $1.8\ \Omega$ respectively. Find the secondary terminal voltage when supplying 10 A at 0.8 p.f. lagging.

(Electric. Machines-II, Bangalore Univ. 1991)

Solution.

$$K = 460/230 = 2; R_{02} = R_2 + K^2 R_1 = 0.75 + 2^2 \times 0.2 = 1.55 \Omega$$

$$X_{02} = X_2 + K^2 X_1 = 1.8 + 2^2 \times 0.5 = 3.8 \Omega$$

$$\text{Voltage drop} = I_2 (R_{02} \cos \phi + X_{02} \sin \phi) = 10 (1.55 \times 0.8 + 3.8 \times 0.6) = 35.2 \text{V}$$

$\therefore$ Secondary terminal voltage = $460 - 35.2 = 424.8 \text{ V}$

Example 32.22. Calculate the regulation of a transformer in which the percentage resistance drops is 1.0% and percentage reactance drop is 5.0% when the power factor is (a) 0.8 lagging (b) unity and (c) 0.8 leading. (Electrical Engineering, Banaras Hindu Univ. 1988)

(Electrical Engineering, Banaras Hindu Univ. 1988)

Soluion. We will use the approximate expression of Art 30.16.

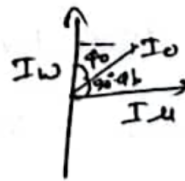
(a) p.f. = $\cos \phi = 0.8$ lag $\mu = v_r \cos \phi + v_x \sin \phi = 1 \times 0.8 + 5 \times 0.6 = \mathbf{3.8\%}$

(b) p.f. = $\cos \phi = 1$ $\mu = 1 \times 1 + 5 \times 0 = 1\%$

(c) $\text{p.f.} = \cos \phi = 0.8 \text{ lead}$ $\mu = 1 \times 0.8 - 5 \times 0.6 = -2.2\%$

Type: 4

$$\begin{aligned}
 32.9 \quad & I_w = I_0 \cos \phi_0 \\
 32.27 \quad & I_\mu = I_0 \sin \phi_0 \\
 32.28(a) \quad & I_0^2 = I_\mu^2 + I_w^2 \\
 & I_0 = \sqrt{I_\mu^2 + I_w^2}
 \end{aligned}$$



- ⑥ No load power $\Rightarrow P_0 = V_0 I_0 \cos \phi_0$
- ⑦ Winding loss, Cu-loss $\rightarrow I_0^2 R$ (No load)
- ⑧ No load power = Core loss + Cu-loss
 $\therefore \text{Core loss} = \text{No load power} - \text{Cu-loss} \Rightarrow P_0 - I_0^2 R$

Example 32.9. (a) A 2,200/200-V transformer draws a no-load primary current of 0.6 A and absorbs 400 watts. Find the magnetising and iron loss currents.

(b) A 2,200/250-V transformer takes 0.5 A at a p.f. of 0.3 on open circuit. Find magnetising and working components of no-load primary current.

Solution. (a) Iron-loss current

$$= \frac{\text{no-load input in watts}}{\text{primary voltage}} = \frac{400}{2,200} = 0.182 \text{ A}$$

Now $I_0^2 = I_w^2 + I_\mu^2$

Magnetising component $I_\mu = \sqrt{(0.6^2 - 0.182^2)} = 0.572 \text{ A}$

The two components are shown in Fig. 29.15.

Example. 32.27. In no-load test of single-phase transformer, the following test data were obtained :

Primary voltage : 220 V ; Secondary voltage : 110 V ;

Primary current : 0.5 A ; Power input : 30 W.

Find the following :

(i) The turns ratio (ii) the magnetising component of no-load current (iii) its working (or loss) component (iv) the iron loss.

Resistance of the primary winding = 0.6 ohm.

Draw the no-load phasor diagram to scale.

(Elect. Machine A.M.I.E. 1990)

Solution. (i) Turn ratio, $N_1/N_2 = 220/110 = 2$

(ii) $W = V_1 I_0 \cos \phi_0$; $\cos \phi_0 = 30/220 \times 0.5 = 0.273$; $\sin \phi_0 = 0.962$

$$I_\mu = I_0 \sin \phi_0 = 0.5 \times 0.962 = 0.48 \text{ A}$$

(iii) $I_w = I_0 \cos \phi_0 = 0.5 \times 0.273 = 0.1365 \text{ A}$

(iv) Primary Cu loss = $I_0^2 R_1 = 0.5^2 \times 0.6 = 0.15 \text{ W}$

$\therefore$ Iron loss = $30 - 0.15 = 29.85 \text{ W}$

Types 5

- ③ Hysteresis loss, $P_h = k_h f B_m^{1.6}$
- ④ Eddy current loss, $P_e = k_e f^2 B_m^2$ } 32.29, 32.32

Example 32.29. In a transformer, the core loss is found to be 52 W at 40 Hz and 90 W at 60 Hz measured at same peak flux density. Compute the hysteresis and eddy current losses at 50 Hz.

(Elect. Machines, Nagpur Univ. 1993)

Solution. Since the flux density is the same in both cases, we can use the relation

$$\text{Total core loss } W_i = Af + Bf^2 \quad \text{or} \quad W_i/f = A + Bf$$

$$\therefore 52/40 = A + 40B \quad \text{and} \quad 90/60 = A + 60B ; \quad \therefore A = 0.9 \quad \text{and} \quad B = 0.01$$

At 50 Hz, the two losses are

$$W_h = A_f = 0.9 \times 50 = 45 \text{ W} ; W_e = Bf^2 = 0.01 \times 50^2 = 25 \text{ W}$$

Example 32.32. When a transformer is connected to a 1000-V, 50-Hz supply the core loss is 1000 W, of which 650 is hysteresis and 350 is eddy current loss. If the applied voltage is raised to 2,000 V and the frequency to 100 Hz, find the new core losses.

Solution. Hysteresis loss $W_h \propto B_{\max}^{1.6} f = PB_{\max}^{1.6} f$

Eddy current loss $W_e \propto B_{\max}^2 f^2 = QB_{\max}^2 f^2$

From the relation $E = 4.44 f N B_{\max}$ A volt, we get $B_{\max} \propto E/f$

Putting this value of $B_{\max}$ in the above equations, we have

$$W_h = P \left(\frac{E}{f} \right)^{1.6} f = P E^{1.6} f^{-0.6} \text{ and } W_e = Q \left(\frac{E}{f} \right)^2 f^2 = Q E^2$$

In the first case,

$$E = 1000 \text{ V}, f = 50 \text{ Hz}, W_h = 650 \text{ W}, W_e = 350 \text{ W}$$

$$\therefore 650 = P \times 1000^{1.6} \times 50^{-0.6} \quad \therefore P = 650 \times 1000^{-1.6} \times 50^{0.6}$$

$$\text{Similarly, } 350 = Q \times 1000^2 \quad \therefore Q = 350 \times 1000^{-2}$$

Hence, constants P and Q are known.

Using them in the second case, we get

$$W_h = (650 \times 1000^{-1.6} \times 50^{0.6}) \times 2000^{1.6} \times 100^{-0.6} = 650 \times 2 = 1,300 \text{ W}$$

$$W_e = (350 \times 1000^{-2}) \times 2,000^2 = 350 \times 4 = 1,400 \text{ W}$$

$\therefore$ Core loss under new condition is $= 1,300 + 1,400 = 2700 \text{ W}$

Alternative Solution

Here, both voltage and frequency are doubled, leaving the flux density unchanged.

With 1000 V at 50 Hz

$$W_h = A f \text{ or } 650 = 50 A ; A = 13$$

$$W_e = B f^2 \text{ or } 350 = B \times 50^2 ; B = 7/50$$

With 2000 V at 100 Hz

$$W_h = A f = 13 \times 100 = 1300 \text{ W and}$$

$$W_e = B f^2 = (7/50) \times 100^2 = 1400 \text{ W}$$

$\therefore$ New core loss $= 1300 + 1400 = 2700 \text{ W}$

Fradays Law

Example: A coil that has 700 turns develops an average induced voltage of 50 V. What must be the change in the magnetic flux that occur to produce such a voltage if the time interval for this change is 0.7 seconds?

Solution:

Given:

Number of Turn (N) = 700

Induced Voltage (ε) = 50 V

Time Interval (dt) = 0.7 s

By using Faraday's Law,

$$\varepsilon = N (d\phi/dt)$$

Therefore, Change in flux (d ϕ) is,

$$d\phi = \varepsilon \times dt / N$$

$$= 50 \times 0.7 / 700$$

$$d\phi = 0.05 \text{ Wb}$$

Example: The magnetic flux linked with a coil having 250 turns is changed from 1.4 Wb to 2 Wb in 0.45 seconds. Calculate the induced emf in the coil.

Solution:

Given:

Number or Turns (N) = 250

Time Interval (dt) = 0.45 s

Initial Flux (ϕ_1) = 1.4 Wb

Final flux (ϕ_2) = 2 Wb

Therefore, change in flux (d ϕ) can be given by,

$$d\phi = \phi_2 - \phi_1 = 2 - 1.4 = 0.6 \text{ Wb}$$

According to Faraday's Law,

$$\text{Induced Emf } (\varepsilon) = N (d\phi/dt)$$

$$\varepsilon = 250 \times (0.6 / 0.45)$$

$$\varepsilon = 333.33 \text{ V}$$